

XIAMEN
UNIVERSITY

COM102 – Programming Language C

Chapter 7: Input and Output (i) – Lab

Dr. Najla'a A. M. Draib

Exercise 1

Write a C program that does the following:

Task 1:

- **Opens** (or creates if it does not exist) a file named **output.txt** in write mode.
- **Writes** a single character **'A'** to the file.
- **Closes** the file properly.

If the file cannot be opened, the program should display an **error message**.

Exercise 1 – Task 1- Solution

```c
#include <stdio.h>

int main() {
    FILE *file;

    // Open (or create) the file in write mode
    file = fopen("output.txt", "w");
    if (file == NULL) {
        printf("Error opening file!\n");
        return 1; // Exit with error code
    }

    // Write a single character 'A' to the file
    fputc('A', file);

    // Close the file
    fclose(file);

    return 0; // Successful execution
}
```

Exercise 1

Task 2:

Modify your program so that, after writing the single character, it also **writes all the elements of the following character array** to the same file:

```
char letters[] = {'H', 'e', 'l', 'l', 'o', '\n'};
```

Exercise 1 – Task 2 Solution

```c
#include <stdio.h>
int main() {
    FILE *file;
    char letters[] = {'H', 'e', 'l', 'l', 'o', '\n'};
    int i;

    // Open (or create) the file in write mode
    file = fopen("output.txt", "w");
    if (file == NULL) {
        printf("Error opening file!\n");
        return 1; // Exit with error code
    }

    // (a) Write a single character 'A' to the file
    fputc('A', file);

    // (b) Write all elements of the array to the file using a loop
    for (i = 0; i < 6; i++) {
        fputc(letters[i], file);
    }

    // Close the file
    fclose(file);

    return 0; // Successful execution
}
```

Exercise 1

Task 3:

- After writing to the file, **reopen** output.txt in read mode and display all the contents of the file on the screen (exactly as stored in the file). If the file cannot be opened for reading, the program should display an error message.

Exercise 1 – Task 3 Solution

```c
int main() {
    FILE *file;
    char letters[] = {'H', 'e', 'l', 'l', 'o', '\n'};
    int i;

    // Open (or create) the file in write mode
    file = fopen("output.txt", "w");
    if (file == NULL) {
        printf("Error opening file!\n");
        return 1; // Exit with error code
    }

    // (a) Write a single character 'A' to the file
    fputc('A', file);

    // (b) Write all elements of the array to the file using a loop
    for (i = 0; i < 6; i++) {
        fputc(letters[i], file);

    }
    fclose(file);
file = fopen("output.txt", "r");
if (file == NULL) {
    printf("Error opening file for reading!\n");
    return 1;
}
char ch;
while ((ch = fgetc(file)) != EOF) {
    putchar(ch);
}


    // Close the file
    fclose(file);

    return 0; // Successful execution
}
```